Portable Records, Recoverable Skills: Equal Switching Gains, Different Product Designs

Zhenyuan Pan
Graduate Student, Department of Economics, Graduate School of Arts and Science,
New York University, New York, NY, USA
Corresponding author: Zhenyuan Pan. Email address: zp2411@nyu.edu (Zhenyuan Pan)

September 2026

Abstract

A user leaving a digital platform may need to move records and recover the ability to use them. This paper asks how these two forms of help change the platform's product design. A two-period duopoly model compares improvements that remove the same switching loss at the existing design. With a fixed recovery fraction, help directed at skill induces a larger specialisation response than an equally effective record export. The comparison also holds for finite changes under stated conditions. Record portability can nevertheless make departure easier while designs become more specialised. Users' own recovery effort changes the comparison. Better recovery tools replace some private practice, and an explicit threshold determines when record coverage produces the larger design response. The model then traces these responses into lifetime surplus before implementation costs. Its central implication is that the immediate gain from switching leaves part of the investment response unexplained. The part of the acquired relationship that an improvement restores determines how firms value further specialisation.

JEL codes: L13, L15, L86, D43.

Keywords: data portability; specific human capital; switching costs; platform design; skill recovery.

1 Introduction

Learning to use a service can make it more productive and harder to leave. Consider a user who has learned how to phrase requests to an AI assistant and interpret its responses. A conversation archive preserves information from that relationship. A tool that helps the user adapt the familiar workflow to another assistant preserves a different asset: the ability to use that information. This example separates what a user can carry away from what the user can do after moving. Studies of web use link practice to greater efficiency (Johnson et al., 2003), while evidence on consumer choice shows how acquired skills can favour a familiar service (Murray and Häubl, 2007).

The asset distinction draws on the economics of specific human capital. Becker's distinction between general and specific training (Becker, 1962) and the skill-combination argument in Lazear

(2009) explain how learning can have value tied to a particular relationship. Here that value resides partly in the user. Wernerfelt (1985) models how experience with a brand develops skills that make the familiar product more useful. This gives a productive basis for loyalty. Models of data-enabled learning study how information accumulated by firms creates competitive advantage (Prüfer and Schottmüller, 2021; Hagiu and Wright, 2023). Separating these locations of learning makes it possible to compare moving information with restoring a user's ability to use it.

A platform has to decide how much of that learning should be specific to its own product. Specific learning can raise output within the relationship, while skills that transfer poorly make departure costly. This trade-off connects productive adaptation to the competition for users. The switching-cost literature shows that future retention rents affect prices offered to new customers (Klemperer, 1987, 1995). Loyalty-related price commitments can create switching costs (Caminal and Matutes, 1990). Firms adjust the frictions they control when external switching costs change (Shi, 2013). The timing of these choices also affects which frictions they create (Rhee, 2014). These findings make product design part of the response to easier switching.

The question here is how the form of switching help affects that response. Compare a larger record export with a better tool for recovering existing skills. Suppose the two improvements remove exactly the same loss at the current product design. A user considering departure initially gains the same amount from either improvement. Does the platform then have the same incentive to make its product more specialised? Holding the initial gain fixed allows the analysis to isolate which part of the relationship the help protects.

This comparison matters because an immediate mobility gain can conceal a change in future design. A recovery tool can preserve the value of additional specific learning, which reduces the cost of specialising further. A record export can also remove information losses that arise even when learning is general. These improvements may therefore look equally effective before firms respond, yet lead to different products afterwards. An evaluation based only on the initial cost of leaving would omit the resulting change in productive learning. The distinction between an export right and its effective use also appears in the implementation discussion of Krämer (2021).

Existing portability models establish that firms respond along several margins. Compatibility choices can change the effect of transferring data (Jeon et al., 2023). Forward-looking demand can also give a firm a reason to promise portability, despite the loss of future retention rents (Hagiu and Wright, 2026). Portability can also change the amount of information firms obtain. When coverage is limited to user-provided data, additional disclosure can reinforce the incumbent's data advantage (Lam and Liu, 2020). Entrants may also respond by collecting more data (Wohlfarth, 2019). The present analysis fixes the initial stocks and compares what is restored from the user's acquired relationship, holding the initial outside-option gain equal.

The composition of a switching cost already matters in earlier models. Nilssen (1992) holds total switching costs constant while varying the balance between recurring transaction costs and the cost of learning a new supplier. That composition changes competition. Here the matched object is the

reduction in the optimised loss at an existing design. Productive specialisation can then adjust, and private recovery effort can reverse the comparison between the two forms of help.

A related distinction arises in the composition of customer information. Colombo et al. (2025) distinguish how precisely firms identify customers from how widely they identify them. Their analysis shows why equal amounts of information can have different pricing consequences. The present comparison asks which design incentives remain different when the initial switching gain is equal. Its asset structure and endogenous recovery give that comparison specific economic restrictions.

The model follows one user relationship through two periods of competition. Each platform chooses the share of learning that is specific to its service. A user who later switches can take part of the record and recover part of the skill's productive value. Platforms set different second-period prices for existing users and users arriving from the rival. Future switching losses consequently affect both the attraction of the relationship today and the rents available tomorrow. This pricing structure turns the difference between the two assets into a difference in design incentives.

With a fixed recovery fraction, skill recovery induces the larger specialisation response. The reason is that its entire benefit comes through specific skill. Record coverage also helps users by preserving information whose value does not depend on specialisation. For the same initial gain, recovery therefore protects more of the next unit of specific learning. Under stated conditions, this comparison holds for finite improvements as well as small ones. There are also parameter values for which greater record coverage reduces specialisation while better skill recovery raises it.

More specialised design can still leave users with a lower cost of departure. When records and skills work better together, taking more records to a rival preserves more of the return to specific learning. The platform can then choose greater specialisation, even as the user loses less from switching. The paper gives sufficient conditions for this combination over the full coverage interval. This result links the productive value of specific learning to effective mobility: a change in product specificity alone is an incomplete account of the change in dependence.

Users' own recovery effort can reverse the comparison between the two kinds of help. A user may spend time learning to apply an existing workflow at the destination. Better recovery tools replace some of that work. More records, in contrast, can make practice more valuable when information complements skill. The model identifies an explicit threshold at which these adjustments reverse the fixed-recovery ordering. The threshold shows when recovery support mainly saves effort and when it also strongly protects the return to additional specific learning.

These design differences also change the gains received over the relationship's lifetime. At the same starting equilibrium and for the same small direct loss reduction, the intervention that induces more specialisation has the larger marginal surplus response before implementation costs. Its additional specific skill can absorb more of the initial mobility gain. A larger productive gain can therefore accompany a smaller final reduction in switching loss. First-period price adjustments determine how much of the gain reaches consumers.

The analysis begins with the value of the acquired assets and the pricing game they support. It then compares the two forms of help, before allowing users to rebuild skill themselves. The

final discussion follows the design response into lifetime surplus and the measurements needed to distinguish the mechanisms.

2 Productive Learning and Competition

2.1 What the user acquires

The platform chooses how much learning becomes specific to its service. There are two platforms, A and B, at the ends of a Hotelling line of length one. A unit mass of users chooses between them in each of two periods. A user's location represents a preference for one service over the other. The cost per unit of distance is $t > 0$, and the common discount factor is $\delta \in (0, 1]$. Locations are uniform and independently redrawn across periods. A sufficiently large common service value keeps the market covered, so every user buys from one platform. Prices are unrestricted.

A new relationship produces both a record and a stock of skill. The record stock is $M > 0$, and total skill is $L > 0$. At date 0, platform j chooses specialisation $s_j \in [0, 1]$ at cost $C(s_j) = k_s s_j^2/2$, where $k_s > 0$. Specific skill is $S_j = s_j L$, and general skill is $(1 - s_j)L$. Fixing M and L isolates the composition of learning. A higher s_j can represent a product built around a platform-specific workflow. A lower value places more weight on transferable procedures.

Accumulated complementary assets can give a familiar supplier a lasting advantage. Huang (2022) studies this mechanism through compatible camera lenses and switching between camera brands. The asset here is the user's acquired skill, whose productive value may be only partly recoverable at a new service. This interpretation gives specialisation a productive benefit as well as a switching penalty.

Specificity affects the value of the relationship as well as the cost of leaving. Output at the original platform is

$$f(s) = q_0 + \beta M + u_G(1 - s)L + u_S sL + \chi MsL. \tag{1}$$

The common service component is q_0. Records have direct value $\beta > 0$. General skill has value $u_G \geq 0$ per unit, while specific skill has value $u_S > 0$. The term χMsL, with $\chi \geq 0$, allows records and specific skill to be complements. For example, a familiar way of querying an assistant may be more useful when the assistant also has the relevant history. This is a technological assumption of the model.

The productive return to specialisation enters both periods. The relationship delivers $\omega f(s)$ in period 1 and $f(s)$ in period 2, where $\omega \geq 0$. It is useful to collect the relevant values as

$$a = L(u_S + \chi M), \quad \gamma = L\chi M, \quad m = \beta M, \qquad A = (\omega + \delta)(a - Lu_G). \tag{2}$$

Here m is the direct value of the record, and a is the value of a unit of specialisation before subtracting displaced general skill. The term γ collects record-skill complementarity. The gross lifetime output return to specialisation is A.

2.2 What the user loses by switching

The user can carry information and recover skill to different degrees. Coverage $\tau \in [0,1]$ is the fraction of the record available at the destination. Recoverability $\rho \in (0,1)$ is the fraction of specific skill's productive value restored there. General skill transfers fully. The second-period loss from departure is

$$D(s,\tau,\rho) = m(1-\tau) + Lu_S(1-\rho)s + \gamma(1-\tau\rho)s = d + bs, \tag{3}$$

where

$$d = m(1-\tau), \qquad b = b_0 - b_1\tau, \qquad b_0 = Lu_S(1-\rho) + \gamma, \quad b_1 = \gamma\rho. \tag{4}$$

The intercept d is the loss that remains even with no specific skill. The slope b measures the additional loss caused by one more unit of specialisation. This separation will matter when two improvements reduce today's loss by the same amount but change its slope differently.

Coverage and recoverability represent two distinct forms of switching help. Both are common external conditions that platforms observe before choosing their designs. A broader export raises τ. A tool that lets the user apply a familiar workflow at the destination raises ρ. Converters can provide compatibility between existing systems while leaving costs or performance losses, as in Farrell and Saloner (1992). The recovery fraction represents the value restored by such help in the present setting. In this benchmark it is fixed; Section 4 allows the user to improve it through effort.

The acquired stock remains tied to the design under which it was formed. A user originally served by j obtains $f(s_j) - D_j$ after switching. The destination supplies the same base service and uses the common transfer technology to restore the acquired skill. Its own specialisation choice shapes the learning of its original users. It does not directly change the incoming user's stock within the second period. This assumption describes a transition in which the immediate task is to reuse an existing workflow. It also makes each origin's loss depend on its own design.

2.3 Why tomorrow's loss affects competition today

Users value the possibility of leaving before they choose where to start. After design, platforms set first-period prices and users choose a relationship. In period 2, platforms observe purchase histories and set separate prices for each history group. Users then decide whether to leave. Second-period service cost is c, and first-period service cost is c_p. In the effort extension, a user who switches also chooses how much skill to rebuild.

Purchase history matters here because switching destroys part of an acquired asset. Independently redrawn preferences follow one case in Fudenberg and Tirole (2000); the real loss D_j sustains a difference between the original supplier and its rival. History-based discounts belong to the broader customer-recognition literature (Chen, 1997; Fudenberg and Villas-Boas, 2006). Past purchases also allow firms to target rivals' customers in the continuing-market model of Villas-Boas (1999). The two-period structure here connects these pricing incentives to the design of the acquired asset. Solving its pricing game gives the lifetime payoff against which a platform chooses specialisation.

Lemma 1 (Lifetime competition)**.** *For a history group with $0 \leq D_j < 3t$, the interior second-period prices and incumbent share are*

$$p_{Ij} = c + t + \frac{D_j}{3}, \qquad p_{Ej} = c + t - \frac{D_j}{3}, \qquad \sigma_j = \frac{1}{2} + \frac{D_j}{6t}. \tag{5}$$

Define the retention and poaching profits per original user by

$$R(D) = \frac{(t + D/3)^2}{2t}, \qquad P(D) = \frac{(t - D/3)^2}{2t}.$$

Let

$$h(D) = -\frac{D}{2} + \frac{5D^2}{36t}, \qquad H_j = As_j + \delta h(D_j). \tag{6}$$

If $|H_A - H_B| < 3t$, first-period shares and reduced profits satisfy

$$n_A = \frac{1}{2} + \frac{H_A - H_B}{6t}, \qquad \Pi_j = 2tn_j^2 + \delta P(D_{-j}) - \frac{k_s s_j^2}{2}. \tag{7}$$

First-period prices are

$$p_{1j} = c_p + 2tn_j - \delta\{R(D_j) - P(D_{-j})\}. \tag{8}$$

A higher switching loss allows the original supplier to charge more in period 2. The rival must offer a lower price to attract the same history group. Yet the initial customer also anticipates these future terms. The first-period price in (8) reflects the value of retaining that customer later. This is the connection between future dependence and current competition emphasised in switching-cost models (Farrell and Klemperer, 2007).

After this initial competition, the rival's customers still generate poaching profits. The term $P(D_{-j})$ in (7) is platform j's profit from those customers. The persistence of such returns after competition for new users also appears in Gehrig and Stenbacka (2004). A unilateral change in s_j leaves this term unchanged. The platform therefore chooses specialisation through its own relationship's attractiveness, H_j, and its design cost.

The function h summarises how switching loss affects that attractiveness. For $D < 9t/5$, $h'(D) < 0$ and $h''(D) = 5/(18t) > 0$. Adding a loss makes the relationship less attractive. The marginal penalty is greater when the existing loss is lower. Easier switching can therefore strengthen the pressure to avoid further dependence. At the same time, an improvement can lower D_s, the extra loss from another unit of specialisation. The design response turns on the balance between these effects.

2.4 The platform's design choice

A well-defined design comparison requires an equilibrium for each environment being compared. The following restrictions keep switching losses moderate and make the cost of design sufficiently curved. They cover all designs in the strategy interval, including unilateral departures from the equilibrium. Let $q = 5/(18t)$; this shorthand differs from the common service value q_0.

Assumption 1 (Moderate losses and sufficient design curvature)**.** For the fixed recovery fraction, the primitives satisfy

$$\begin{aligned} m + b_0 &< \frac{9t}{5}, & A &< 3t, \\ A &> \frac{\delta b_0}{2}, & A &< 3k_s, \\ k_s &> \frac{2A^2}{9t} + \frac{2\delta q b_0^2}{3}. \end{aligned} \tag{9}$$

The loss bound keeps both suppliers active in second-period competition. The bounds on A ensure a positive productive incentive without pushing the platform to either design endpoint. The cost restriction limits how strongly the platforms amplify each other's design incentives. These are sufficient conditions for the equilibrium used below. Table 1 provides one parameter set satisfying them.

Proposition 1 (Unique equilibrium)**.** *Under Assumption 1, for every $\tau \in [0,1]$, the reduced design game has a unique pure-strategy equilibrium. It is symmetric and interior. Together with the pricing strategies in Lemma 1, it gives a subgame-perfect equilibrium. Let*

$$K = 3k_s - \delta q b^2 > 0. \tag{10}$$

Then

$$s^* = \frac{A + \delta b(-1/2 + qd)}{K}, \qquad D^* = \frac{3k_s d + Ab - \delta b^2/2}{K}. \tag{11}$$

The uniform claim concerns coverage at fixed ρ. Small changes in ρ are admissible wherever the strict inequalities continue to hold.

The equilibrium follows from a common function that tracks either platform's incentives. This function is an exact potential in the sense of Monderer and Shapley (1996):

$$\Phi(s_A, s_B) = \frac{H_A + H_B}{3} + \frac{(H_A - H_B)^2}{18t} - \frac{k_s}{2}(s_A^2 + s_B^2). \tag{12}$$

A unilateral design change raises the platform's profit by exactly the change in Φ. Assumption 1 makes Φ strictly concave on the full strategy square, so it has one maximiser. Every design equilibrium is that maximiser. Symmetry then gives the common choice in (11). The Online Appendix establishes these steps and the pricing bounds.

The economic trade-off appears directly in the symmetric design condition:

$$A + \delta h'(D) D_s = 3k_s s. \tag{13}$$

An additional unit of specialisation raises lifetime output by A. It also raises the switching loss by D_s, which changes the relationship's attraction by $\delta h'(D) D_s$. The platform balances the resulting return against design cost. An improvement that changes the size of the loss need not change the cost of additional specialisation in the same way. That is the source of the comparison that follows.

3 Two Ways to Make Switching Easier

3.1 Comparing equal gains at the existing design

Equal initial help can protect different parts of the next design. Return to the example of an exported conversation record and a tool for recovering a familiar workflow. Suppose each removes the same loss at the design users have already learned. The platform now considers making that design a little more specific. Its response depends on how much of the additional switching loss each tool would remove. Matching the initial gain isolates this second question.

The comparison uses derivatives to distinguish the amount of help from where it acts. Let $D(s, z)$ be a twice differentiable effective loss near a symmetric interior equilibrium, with $D_s > 0$ and $D_z < 0$. Here z denotes an improvement, and D_z measures its effect while the design stays fixed. The loss may already include optimally chosen recovery effort. Allow gross lifetime output to be $V(s)$ and design cost to be $C(s)$, both independent of z. The symmetric condition is $V_s + \delta h'(D) D_s = 3C'(s)$. Define

$$\mathcal{K} = 3C'' - V_{ss} - \delta\{h'' D_s^2 + h' D_{ss}\} > 0, \qquad \ell_z = -D_z > 0. \tag{14}$$

The positive curvature term $\mathcal{K}$ makes the local design response well-defined. Direct relief is $\ell_z = -D_z$. Dividing the equilibrium response by this amount gives $\mathcal{R}_z = s_z^*/\ell_z$, the change in specialisation per unit of initial loss reduction. This places differently measured interventions on the same scale.

The remaining statistic measures how that help changes with specialisation. A larger value means that the improvement removes more of the extra loss attached to specific learning, relative to the direct relief it already provides. Theorem 1 formalises this comparison.

Theorem 1 (Matched-relief comparison)**.** *At a symmetric interior equilibrium satisfying* (14)*, with* $h'(D) < 0$*, define*

$$\eta_z = \frac{-D_{sz}}{-D_z}. \tag{15}$$

Then

$$\mathcal{R}_z = \frac{\delta}{\mathcal{K}} \left[(-h')\eta_z - h'' D_s \right]. \tag{16}$$

For two interventions z_1, z_2 *that each produce the same infinitesimal fixed-design relief* $\mathrm{d}\varepsilon$*,*

$$\frac{\mathrm{d}s^*|_{z_2} - \mathrm{d}s^*|_{z_1}}{\mathrm{d}\varepsilon} = \frac{\delta(-h')}{\mathcal{K}} (\eta_{z_2} - \eta_{z_1}). \tag{17}$$

Their different design responses depend only on the difference in design sensitivity, conditional on the starting equilibrium.

Proof. Differentiate the symmetric design condition. This gives $\mathcal{K} s_z^* = \delta\{h'' D_z D_s + h' D_{sz}\}$. Divide by $\ell_z = -D_z$ to obtain (16), then subtract the two equations. The common level effect $-\delta h'' D_s/\mathcal{K}$ cancels. □

Matching removes the competitive effect common to both improvements. The term $-\delta h'' D_s/\mathcal{K}$

captures the stronger penalty for adding a switching loss after the initial loss has fallen. Equal direct relief makes that effect the same. The remaining difference comes from η_z: the protection offered to the marginal unit of specific learning. This is why the same initial change in prices and retention can produce different designs.

The statistic has a practical interpretation in terms of a loss schedule. Imagine measuring how much each tool helps at a less specialised design and at a more specialised design. The proportional change in that help is $\eta_z = \partial_s \log \ell_z$. The statistic is unchanged if the instrument is expressed in different units. A negative value is possible when help is concentrated on less specialised designs. The asset structure below determines the values relevant to records and skills.

3.2 Why skill recovery changes the design incentive more

A record export has value even when all learning is general. Skill recovery becomes useful through the specific skill that would otherwise lose value at the destination. This difference places more of the recovery tool's benefit on the margin the platform controls. With $c_\tau = L(u_S + \chi\tau M)$, the benchmark gives

$$\ell_\tau = m + b_1 s, \qquad \ell_\rho = c_\tau s, \qquad \eta_\tau = \frac{b_1}{m + b_1 s}, \qquad \eta_\rho = \frac{1}{s}. \tag{18}$$

The direct record value m appears in coverage relief, but contributes nothing to the relief from additional specialisation. Skill recovery has no corresponding design-independent component. After matching the total initial gain, recovery therefore reduces the marginal switching penalty more strongly.

Corollary 1 (Asset-specific design responses)**.** *Under Assumption 1, at an interior coverage and recovery pair,*

$$\mathcal{R}_\rho - \mathcal{R}_\tau = \frac{\delta[-h'(D^*)]m}{Ks^*(m + b_1 s^*)} > 0. \tag{19}$$

When $\chi = 0$, increasing record coverage reduces specialisation. Improving skill recovery increases specialisation whenever $d + 2bs^* < 9t/5$.

This result compares the responses, not necessarily two increases. The larger response may be a smaller decline. At $\chi = 0$, however, the model gives a particularly simple opposite-sign case. Record coverage removes an additive information loss, which strengthens the competitive penalty for adding further dependence. Its design effect is $s^*_\tau = -\delta qbm/K < 0$. Better skill recovery also lowers the loss created by the next unit of specialisation. Its response is positive when $d + 2bs^* < 9t/5$.

The direct value of records explains the strict gap between the responses. When $m = 0$ and both instruments still provide positive relief, the difference in (19) is zero. All record benefits then operate through specific skill, just as recovery benefits do. Complementarity determines whether greater coverage itself encourages specialisation; positive direct record value separates coverage from recovery in the matched comparison.

3.3 The difference also holds for finite improvements

A finite comparison requires the two improvements to be matched at one common design. Start from (τ_0, ρ_0) and its equilibrium s_0. Choose a feasible loss reduction $\varepsilon > 0$. The two new environments are

$$(\tau_R, \rho_R) = \left(\tau_0 + \frac{\varepsilon}{m + b_{1,0} s_0}, \rho_0\right), \qquad (\tau_S, \rho_S) = \left(\tau_0, \rho_0 + \frac{\varepsilon}{c_{\tau_0} s_0}\right). \tag{20}$$

Each reduces $D(s_0, \tau_0, \rho_0)$ by exactly ε. Feasibility requires both new fractions to be at most one. Firms then choose their designs separately in the two environments.

Proposition 2 (Finite design comparison)**.** *Suppose Assumption 1 holds at ρ_0, and $m + 2b_{0,0} < 9t/5$. For every feasible positive matched change in* (20)*, the skill-recovery environment has strictly higher equilibrium specialisation:*

$$s_S^* > s_R^*.$$

The losses are matched at the original design. They need not be equal at the redesigned equilibria.

The two new loss schedules meet at the original design, but have different slopes. Skill recovery produces the flatter schedule because its entire benefit protects specific learning. The additional bound in Proposition 2 makes the resulting difference in marginal design incentives positive throughout the strategy interval. Both new environments retain a unique equilibrium, so the comparison extends beyond a local derivative.

Figure 1 shows this difference as the common initial gain grows. The horizontal axis measures the same loss reduction at s_0 for both tools. The vertical axis shows the designs chosen afterwards. The recovery curve lies above the coverage curve at every positive matched gain in the plotted range. The curves illustrate the finite result using the parameters reported in Table 1.

The comparison leaves a further question about record coverage itself. Even when recovery produces the larger response, coverage may encourage more specialised design. The next result asks whether users can still find it easier to leave after that response.

3.4 More specific design can accompany easier departure

Portable records can preserve the value of specific learning after departure. When a familiar workflow works better with the user's history, a richer export raises the value of using that workflow at a rival. This lowers the extra loss created by specialisation. The platform may then choose a more specific design. Differentiating its equilibrium choice gives

$$\frac{\mathrm{d}s^*}{\mathrm{d}\tau} = \frac{\delta}{K}\left\{b_1\left(\frac{1}{2} - qD^*\right) - qb(m + b_1 s^*)\right\}. \tag{21}$$

The positive component captures protection for additional specific skill. The negative component captures the stronger competitive penalty for adding dependence after the initial loss falls. Complementarity activates the first component through b_1. It also changes the level of productive value and switching losses, so the balance depends on the stated parameter restrictions.

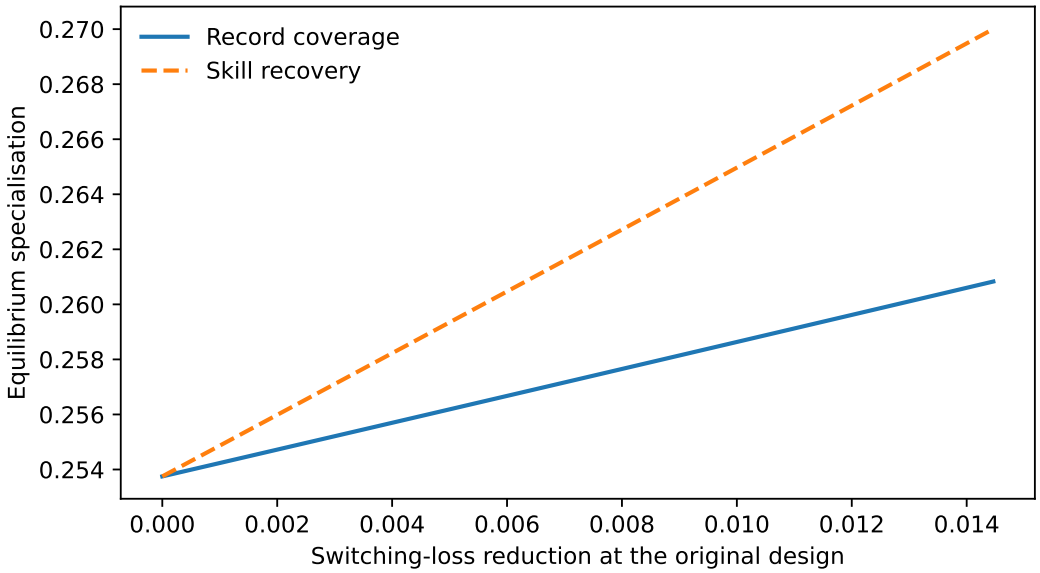


Figure 1: Design responses to the same fixed-design switching gain. Coverage or recovery is increased from $(\tau_0, \rho_0) = (0.5, 0.8)$ so that the loss at the original equilibrium design falls by the amount on the horizontal axis. Curves show newly chosen specialisation. Parameters are in Table 1; all points satisfy the conditions of Proposition 2.

For users, the final loss includes the platform's design adjustment. Greater coverage directly lowers that loss, while induced specialisation can raise it. The following conditions ensure that coverage both encourages specialisation and leaves users with less loss overall. Define $K_0 = 3k_s - \delta q b_0^2$.

Proposition 3 (Uniform coexistence)**.** *Suppose Assumption 1 holds and*

$$b_1\left(\frac{9t}{5} - m - 2b_0\right) > b_0 m, \qquad K_0 > \frac{\delta b_0 b_1}{2m}. \tag{22}$$

Then, for every coverage rate, equilibrium specialisation increases with coverage and the equilibrium switching loss decreases:

$$\frac{\mathrm{d}s^*}{\mathrm{d}\tau} > 0, \qquad \frac{\mathrm{d}D^*}{\mathrm{d}\tau} < 0.$$

Derivatives at the endpoints have their one-sided interpretation.

The first condition makes the protection for additional specific skill strong enough to encourage specialisation. The second limits how much the resulting design change can absorb the direct mobility gain. Both inequalities are strict, so nearby parameter values preserve the result. The Online Appendix proves the comparison over the full coverage interval.

Table 1 shows one such model economy. Moving from zero to full coverage raises specialisation from 0.2322 to 0.2763. The switching loss falls from 0.1057 to 0.0221. These are theoretical examples in model units. The derivatives have the same signs throughout the interval under Proposition 3.

The platform can specialise more because departure destroys less of the return to that choice. Greater specificity therefore coexists with less effective dependence in this parameter region. This

Table 1: More specialisation with a lower switching loss

Coverage τ	s^*	D^*	$\mathrm{d}s^*/\mathrm{d}\tau$	$\mathrm{d}D^*/\mathrm{d}\tau$
0.0	0.232229	0.105735	+0.042098	-0.077053
0.5	0.253749	0.065600	+0.044023	-0.083556
1.0	0.276295	0.022104	+0.046204	-0.090511

Parameters. $t = 1$, $\delta = 0.9$, $\omega = 0.6$, $M = L = 1$, $\beta = 0.05$, $u_S = 0.2$, $u_G = 0.1$, $\chi = 0.2$, $\rho = 0.8$, and $k_s = 0.5$. Thus $A = 0.45$, $b_0 = 0.24$, $b_1 = 0.16$. All sufficient inequalities in Assumption 1 and Proposition 3 have strictly positive margins. Values are model units.

mechanism still leaves recovery itself to be explained. A user can spend time restoring skill, and that effort changes how much each form of help protects the next unit of specialisation.

4 When Users Rebuild Skill

4.1 Recovery tools can replace the user's own effort

A user may restore part of a familiar workflow through practice. For example, the user may try different ways of carrying out the same task at the destination. A better recovery tool can replace some of that work. More exported history can instead make the work more rewarding when records and skills are complements. These adjustments change what the two improvements accomplish before the platform chooses a new design.

External aids change the work a user undertakes. The review by Risko and Gilbert (2016) describes cognitive offloading through actions that reduce a task's internal cognitive demands. In reminder-setting experiments, Chiu and Gilbert (2024) find that greater physical effort in setting reminders reduces their use. These findings motivate making the user's allocation of effort explicit. In the model, assistance and private practice both restore part of an acquired skill.

Let a switching user choose recovery effort $u \geq 0$. The stocks M and L remain fixed, while common recovery infrastructure $r \geq 0$ is exogenous. The fraction of specific skill restored at the destination is

$$\rho(u, r) = 1 - \alpha \exp(-\kappa_r r - \kappa_u u), \qquad 0 < \alpha < 1, \quad \kappa_r, \kappa_u > 0. \tag{23}$$

The parameter α is the share initially unavailable without infrastructure or effort. The parameters κ_r and κ_u measure how effectively the two inputs restore that share. Their exponential form gives diminishing returns. Effort costs $k_U u^2/2 + \ell u$, with $k_U > 0$ and $\ell \geq 0$.

The user balances the recovered output against the time and resources required. The resulting effective loss is

$$D(s, \tau, r) = \min_{u \geq 0} \left\{ m(1 - \tau) + as - sc_\tau \rho(u, r) + \frac{k_U u^2}{2} + \ell u \right\}. \tag{24}$$

Only users who switch pay the recovery cost. The optimised value D therefore includes the residual

output shortfall and the cost of reducing it. It remains the disadvantage relevant for the pricing game in Lemma 1.

An interior effort choice equates marginal recovery benefit with marginal cost. Strict convexity gives a unique optimum, with

$$k_U u + \ell = s c_\tau \rho_u, \qquad Q = k_U + s c_\tau \kappa_u \rho_u > 0. \tag{25}$$

The term Q is the curvature of the effort problem. A larger value makes effort respond less to a change in its return. Differentiation gives

$$u_\tau = \frac{s\gamma\rho_u}{Q} \geq 0, \qquad u_r = -\frac{s c_\tau \kappa_u \rho_r}{Q} < 0. \tag{26}$$

Coverage raises effort when records complement specific skill. The user has more information with which the restored skill can work. Better infrastructure lowers effort because it performs part of the recovery task itself. The user still gains from the tool. Saved work accounts for part of that gain, which limits the additional output restored by the improvement.

4.2 The design equilibrium with recovery effort

The platform anticipates the user's effort when it chooses specialisation. This adds curvature to the effective loss: a user with more specific skill has more reason to rebuild it. That adjustment can strengthen the design response. A sufficient bound on this additional curvature lets the model retain a unique equilibrium. Set

$$B = \delta \left(q a^2 + \frac{a^2 \kappa_u^2}{2 k_U} \right). \tag{27}$$

The bound combines the pricing curvature with the sensitivity of effort to its return. A higher effort-cost curvature k_U reduces the second term. The design-cost condition below then limits the resulting amplification of platform incentives.

Proposition 4 (Recovery-model equilibrium)**.** *Suppose*

$$m + a < \frac{9t}{5}, \qquad \frac{\delta a}{2} < A < \min\{3t, 3k_s\}, \qquad k_s > \frac{2A^2}{9t} + \frac{2B}{3}. \tag{28}$$

For every $\tau \in [0,1]$ and finite $r \geq 0$, the recovery model has a unique symmetric interior design equilibrium. User effort is uniquely optimal and may be zero. The pricing formulas are valid throughout the design strategy domain. Where equilibrium effort is strictly positive, the equilibrium is differentiable in the interventions and $\mathcal{K} > 0$.

The same potential argument applies because each user's loss depends on the origin's design. Recovery effort makes the effective loss concave in specialisation: additional specific skill encourages the user to rebuild more of it. The value and its first derivative remain continuous when effort changes from zero to positive. The Online Appendix bounds the curvature on both sides of that threshold

and establishes a unique design equilibrium. The next comparison concerns its positive-effort region, where the required cross derivatives exist.

4.3 When effort reverses the comparison

Private effort changes how much help reaches the marginal unit of specific skill. Infrastructure now replaces some recovery the user would have undertaken anyway. Coverage raises the return to that recovery when records complement skill. The first derivatives of the optimised loss follow the envelope argument at differentiable points (Milgrom and Segal, 2002). The cross derivatives also include the user's effort response, which changes how the benefit varies with specialisation. Applying these derivatives gives the two design sensitivities:

$$\eta_r = \frac{k_U}{sQ}, \qquad \eta_\tau = \frac{\gamma\left(\rho + sc_\tau\rho_u^2/Q\right)}{m + s\gamma\rho}. \tag{29}$$

The infrastructure sensitivity falls below its fixed-recovery value $1/s$ because $Q > k_U$. The coverage sensitivity includes an extra term from induced practice. These adjustments work in opposite directions relative to the benchmark, so their size determines whether the original comparison survives.

Theorem 2 (Recovery-effort boundary)**.** *At a positive-effort equilibrium satisfying* (28)*, define*

$$\Xi = k_U m - s^2 c_\tau \gamma \kappa_u^2 (1-\rho). \tag{30}$$

For equal infinitesimal fixed-design reductions in the optimised loss,

$$\operatorname{sgn}(\mathcal{R}_r - \mathcal{R}_\tau) = \operatorname{sgn}\Xi. \tag{31}$$

When $\Xi > 0$, recovery infrastructure generates the larger specialisation response. When $\Xi < 0$, record coverage generates the larger response. Both cases occur under the global equilibrium conditions. With $\chi = 0$ and $m > 0$, $\Xi > 0$.

Proof. Subtract the expressions in (29). Using $Q - k_U = sc_\tau\kappa_u\rho_u$ and $\rho_u = \kappa_u(1-\rho)$ gives

$$\eta_r - \eta_\tau = \frac{k_U m - s^2 c_\tau \gamma \kappa_u \rho_u}{sQ(m + s\gamma\rho)}. \tag{32}$$

The denominator is positive. Theorem 1 gives the sign comparison. The Online Appendix verifies primitive conditions and strict bounds for the examples on both sides of the threshold. □

The threshold separates direct record value from the effect of private recovery. The term $k_U m$ favours the fixed-recovery ordering. A valuable direct record component leaves more of coverage's benefit independent of specialisation, while costly effort adjustment limits substitution. The second term depends on the specific skill at stake and its complementarity with records. When that term

Table 2: Recovery effort can reverse the design comparison

m	s^*	u^*	$\mathcal{R}_\tau$	$\mathcal{R}_r$	Ξ
0.050	0.219621	0.077845	0.335313	1.192439	+0.023246
0.001	0.218434	0.077445	1.389991	1.218097	-0.001235

Parameters. The production and design primitives match Table 1, except $\beta = m$ varies and recovery is endogenous. Set $\alpha = 0.8$, $\kappa_u = \kappa_r = 1$, $k_U = 0.5$, $\ell = 0.001$, $\tau = 0.5$, and $r = 0.2$. The normalisation uses the optimised loss at a fixed design, so it already includes the user's effort adjustment. Both games satisfy Proposition 4.

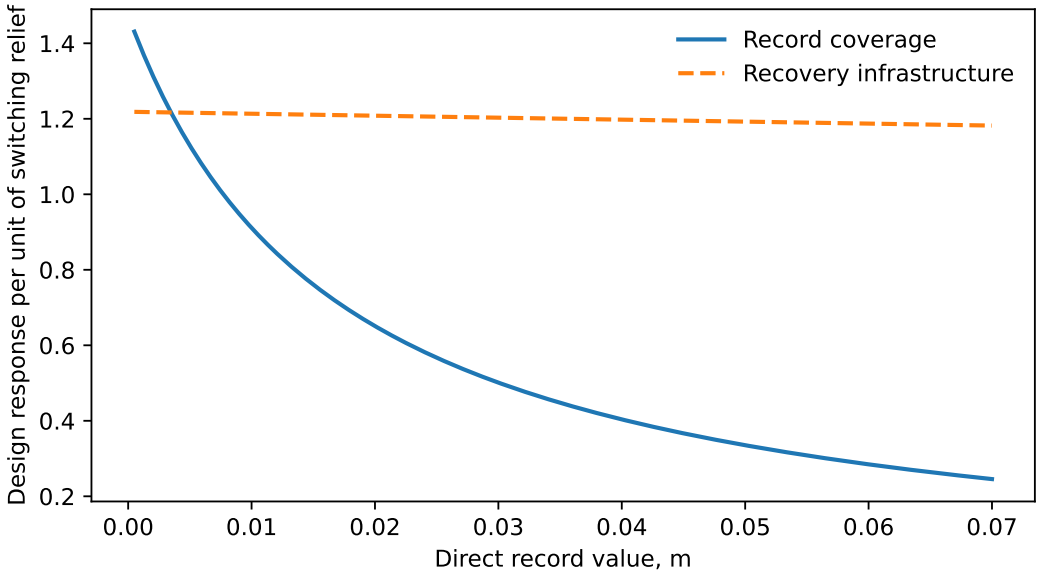


Figure 2: Endogenous effort changes the matched-relief comparison. Each curve gives the specialisation response per unit of fixed-design reduction in the optimised switching loss. The horizontal axis varies m while maintaining the other primitives in Table 2. All plotted games satisfy Proposition 4; the crossing is consistent with $\Xi = 0$.

is larger, coverage gives the stronger design response per unit of direct relief. Both s and ρ are evaluated at the starting equilibrium.

Table 2 illustrates the two sides of this threshold. The production and recovery technologies are held fixed, while the direct record value m changes. At $m = 0.05$, infrastructure generates the larger normalised specialisation response. At $m = 0.001$, coverage generates the larger response. The examples satisfy Proposition 4, and the Online Appendix gives strict analytical bounds for the sign of Ξ.

Figure 2 follows the comparison as direct record value varies. Its vertical axis divides each design response by that intervention's own fixed-design reduction in the optimised loss. Equal increments of coverage and infrastructure would be a different comparison. Matching the direct loss reduction holds the user's initial gain constant after recovery effort has already adjusted.

Observed practice time therefore captures only part of effective mobility. A richer record can induce more practice while lowering the total loss from departure. Better infrastructure can reduce practice while also lowering that loss. Measuring the remaining output shortfall together with the

effort spent restoring it is necessary to tell these mechanisms apart. Their design consequences then determine how the initial gain changes over the relationship's lifetime.

5 What the Design Response Means

5.1 Productivity gains and lifetime prices

An initial switching gain affects both real resources and the prices users pay. The design response changes productive output, while easier departure changes competition for the relationship. To keep these effects separate, let W denote total surplus and CS consumer surplus over both periods. Total surplus counts output net of real costs. Consumer surplus subtracts industry profit from that amount.

Switching uses resources through the remaining output loss and the effort to recover skill. At a symmetric allocation the retained share is $\sigma = 1/2 + D/(6t)$. The real second-period cost from locations and switching is

$$\frac{t}{2}\{\sigma^2 + (1-\sigma)^2\} + (1-\sigma)D = \frac{t}{4} + \frac{D}{2} - \frac{5D^2}{36t}.$$

When recovery effort is endogenous, its cost is already included in D. Up to constants independent of design and interventions, lifetime surplus is therefore

$$W = As + \delta h(D) - k_s s^2, \qquad \text{CS} = As + \delta j(D), \quad j(D) = \frac{D}{6} + \frac{D^2}{36t}. \tag{33}$$

The cost of both platforms' designs enters W. Prices transfer value between users and firms. The Online Appendix derives the accounting expressions with all constants retained.

Lower future retention rents also mean smaller introductory discounts. At symmetry the first-period price is $p_1 = c_p + t - 2\delta D/3$. An anticipated reduction in D raises this price, which offsets part of the gain from easier departure. The fixed-design consumer-surplus effect is negative in this pricing environment. Portability can similarly affect competition for initial users in Hagiu and Wright (2026); Hagiu and Wright (2023) show how ongoing data sharing weakens acquisition subsidies. Here the additional question is how two matched improvements differ once the product is redesigned.

For a small matched improvement, the direct resource gain is common to both instruments. Their surplus difference comes from the different designs that follow. Proposition 5 states this local comparison before implementation costs.

Proposition 5 (Incidence of a matched improvement)**.** *At a symmetric interior equilibrium of either model, for an intervention with $\ell_z > 0$,*

$$\frac{W_z^*}{\ell_z} = \delta(-h') + k_s s^* \mathcal{R}_z, \qquad \frac{\mathrm{CS}_z^*}{\ell_z} = -\delta j' + \left(A + \delta j' D_s\right) \mathcal{R}_z. \tag{34}$$

Consequently the difference in marginal total surplus under equal direct relief is $k_s s^*(\mathcal{R}_{z_2} - \mathcal{R}_{z_1})$. The difference in marginal consumer surplus is $(A+\delta j' D_s)(\mathcal{R}_{z_2} - \mathcal{R}_{z_1})$. Both coefficients are positive under the maintained conditions. Meanwhile,

$$\frac{\mathrm{d}D^*|_z}{\mathrm{d}\varepsilon} = -1 + D_s \mathcal{R}_z. \tag{35}$$

These comparisons exclude implementation costs.

Additional specialisation has a positive marginal social value at the private optimum. The private design condition gives $W_s = H_s - 2k_s s = k_s s$. Thus the intervention producing the larger specialisation response produces the larger marginal total-surplus response under equal direct relief. The consumer-surplus difference has the same sign because its design coefficient is also positive. These are comparisons of local changes; either instrument's absolute consumer-surplus change may still be negative.

More productive design can absorb more of the initial reduction in switching loss. Equation (35) separates the common direct reduction from the loss added by redesign. An intervention can consequently deliver a smaller final mobility improvement and a larger surplus improvement. The finite result in Proposition 2 compares designs only. A finite surplus comparison would also have to account for how these marginal values change along the path.

5.2 What would need to be measured

The mechanism requires a measure of usable mobility, beyond the size of an export. The institutional distinction appears in existing data-portability provisions. Article 20 of the General Data Protection Regulation establishes a qualified right over covered personal data provided by the person. Article 6(9) of the Digital Markets Act addresses end-user data portability for designated gatekeepers' relevant core platform services, including continuous and real-time access (European Parliament and Council of the European Union, 2016, 2022). The model's coverage parameter represents the amount of relevant information transferred. Its recovery parameter represents the ability to use acquired skill at the destination.

Recovery time is itself part of the cost to be measured. Burnham et al. (2003) classify time and effort losses as procedural switching costs. The present loss measure combines those resources with the productivity that remains unavailable after recovery. An empirical application would first measure their combined reduction at a fixed design. In the running example, that means comparing the output shortfall and recovery effort after a larger export with those after better workflow support. The next measurement is how that gain changes with the user's platform-specific skill. Within the

model, $\eta_z = \partial_s \log \ell_z$ links this variation to the design response. The relationship requires that the intervention leave the productive value of staying directly unchanged.

Retention alone combines the loss of skill with price incentives. Evidence from other markets shows why pricing arrangements belong in the interpretation. Viard (2007) finds price declines after 800-number portability in a uniform-pricing setting. A calibrated packaged-goods model in Dubé et al. (2009) instead produces lower prices at moderate switching costs. The present mechanism concerns the separate question of how the component restored by an intervention changes productive design. Its application calls for measures of post-switch productivity and recovery effort alongside observed retention.

5.3 Scope and limitations

The results concern a specific transition between services. The stocks are formed under the origin's design, and the destination applies a common recovery technology to them. This assumption is most natural over a horizon focused on reusing an acquired workflow. A setting in which the destination immediately teaches a new workflow would require its design to enter the incoming user's output. Strategic choices of recovery infrastructure would also introduce cross-platform effects that the current potential does not cover.

Portability may change other parts of the relationship as well. Krämer and Stüdlein (2019) study how portability can weaken an incumbent's incentive to preserve user privacy. Kim (2026) incorporates network benefits and data-breach risks into platform portability and interoperability. Those payoff channels are held fixed here, so the design comparison isolates the productive value of the assets being restored.

The scalar model holds the amount of learning fixed. If training scale can adjust, the two instruments can also change how much skill users acquire. The Online Appendix derives the additional training-feedback term. A strict scalar comparison survives sufficiently small feedback on a regular equilibrium branch. Large feedback can change the ordering, and the scalar uniqueness results do not establish uniqueness for every training technology.

Lifetime incidence depends on how firms can price the relationship. Users anticipate future prices. The market is covered, and preferences are independently redrawn across periods. Prices can adjust freely at the initial date. Nonnegative-price restrictions can change portability outcomes (Jeon and Menicucci, 2024); pricing commitments also matter when preferences persist (Chen and Pearcy, 2010). These alternatives require their own pricing analysis. Likewise, an unexpected change after the initial price has been paid has a different incidence from the anticipated improvements studied here.

Implementation costs must be added before comparing instruments on a resource basis. If intervention z costs Γ_z at the margin, its normalised total-surplus change is reduced by Γ_z/ℓ_z. Consumer incidence also depends on who finances that cost. The paper supplies the design channel before these implementation costs. All numerical values are theoretical constructions. The assistant

example illustrates the declared technology; the paper contains no estimates of skill transfer between AI services or evidence that a particular tool meets the model's conditions.

6 Conclusion

The way switching becomes easier can change the products firms choose to offer. This paper compares moving a user's records with restoring the user's platform-specific skill. The two improvements are matched by the loss they remove at the existing design. Their common competitive effect cancels, which leaves the difference in protection for additional specific learning.

With a fixed recovery fraction, skill recovery produces the larger specialisation response. Record portability can still encourage more specific design while leaving users with a smaller switching loss. When users can practise at the destination, recovery tools replace part of their own effort. The resulting threshold identifies when that substitution reverses the comparison between the two forms of help.

An assessment of mobility therefore needs to connect the user's initial gain to the firm's next design. Measuring the immediate loss reduction describes what the user can recover today. Measuring how that benefit varies with specific skill explains which designs become more attractive tomorrow. Together, these objects link usable portability to productive investment and the surplus generated over the relationship.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT (OpenAI) in order to improve English grammar and assist with LaTeX formatting. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

Becker, G. S. (1962). Investment in human capital: A theoretical analysis. *Journal of Political Economy, 70*(5, Part 2), 9-49. https://doi.org/10.1086/258724

Burnham, T. A., Frels, J. K., & Mahajan, V. (2003). Consumer switching costs: A typology, antecedents, and consequences. *Journal of the Academy of Marketing Science, 31*(2), 109-126. https://doi.org/10.1177/0092070302250897

Caminal, R., & Matutes, C. (1990). Endogenous switching costs in a duopoly model. *International Journal of Industrial Organization, 8*(3), 353-373. https://doi.org/10.1016/0167-7187(90)90002-I

Chen, Y. (1997). Paying customers to switch. *Journal of Economics & Management Strategy, 6*(4), 877-897. https://doi.org/10.1111/j.1430-9134.1997.00877.x

Chen, Y., & Pearcy, J. (2010). Dynamic pricing: When to entice brand switching and when to reward consumer loyalty. *RAND Journal of Economics, 41*(4), 674-685. https://doi.org/10.1111/j.1756-2171.2010.00116.x

Chiu, G., & Gilbert, S. J. (2024). Influence of the physical effort of reminder-setting on strategic offloading of delayed intentions. *Quarterly Journal of Experimental Psychology, 77*(6), 1295-1311. https://doi.org/10.1177/17470218231199977

Colombo, S., Graziano, C., & Pignataro, A. (2025). Personalized pricing with imperfect customer recognition. *Information Economics and Policy, 70*, Article 101129. https://doi.org/10.1016/j.infoecopol.2025.101129

Dubé, J.-P., Hitsch, G. J., & Rossi, P. E. (2009). Do switching costs make markets less competitive? *Journal of Marketing Research, 46*(4), 435-445. https://doi.org/10.1509/jmkr.46.4.435

European Parliament and Council of the European Union (2016). *Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data, and Repealing Directive 95/46/EC (General Data Protection Regulation)*. Official Journal of the European Union, L 119, 4 May 2016, pp. 1-88. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32016R0679

European Parliament and Council of the European Union (2022). *Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on Contestable and Fair Markets in the Digital Sector and Amending Directives (EU) 2019/1937 and (EU) 2020/1828 (Digital Markets Act)*. Official Journal of the European Union, L 265, 12 October 2022, pp. 1-66. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32022R1925

Farrell, J., & Klemperer, P. (2007). Coordination and lock-in: Competition with switching costs and network effects. In M. Armstrong & R. H. Porter (Eds.), *Handbook of Industrial Organization* (Vol. 3, pp. 1967-2072). Elsevier. https://doi.org/10.1016/S1573-448X(06)03031-7

Farrell, J., & Saloner, G. (1992). Converters, compatibility, and the control of interfaces. *Journal of Industrial Economics, 40*(1), 9-35. https://doi.org/10.2307/2950625

Fudenberg, D., & Tirole, J. (2000). Customer poaching and brand switching. *RAND Journal of Economics, 31*(4), 634-657. https://doi.org/10.2307/2696352

Fudenberg, D., & Villas-Boas, J. M. (2006). Behavior-based price discrimination and customer recognition. In T. Hendershott (Ed.), *Economics and Information Systems* (Vol. 1, pp. 377-436). Elsevier. https://doi.org/10.1016/S1574-0145(06)01007-5

Gehrig, T., & Stenbacka, R. (2004). Differentiation-induced switching costs and poaching. *Journal of Economics & Management Strategy, 13*(4), 635-655. https://doi.org/10.1111/j.1430-9134.2004.00026.x

Hagiu, A., & Wright, J. (2023). Data-enabled learning, network effects, and competitive advantage. *RAND Journal of Economics, 54*(4), 638-667. https://doi.org/10.1111/1756-2171.12453

Hagiu, A., & Wright, J. (2026). *Personalization, data portability and competitive dynamics* [Working paper, May 2026]. https://www.andreihagiu.com/Endogenous_switching_costs_20260521_MainPaper.pdf

Huang, Y. (2022). Tied goods and consumer switching costs. *Marketing Science, 41*(1), 93-114. https://doi.org/10.1287/mksc.2021.1312

Jeon, D.-S., & Menicucci, D. (2024). Data portability and competition: Can data portability increase both consumer surplus and profits? *European Journal of Law and Economics, 57*(1-2), 145-162. https://doi.org/10.1007/s10657-023-09774-9

Jeon, D.-S., Menicucci, D., & Nasr, N. (2023). Compatibility choices, switching costs, and data portability. *American Economic Journal: Microeconomics, 15*(1), 30-73. https://doi.org/10.1257/mic.20200309

Johnson, E. J., Bellman, S., & Lohse, G. L. (2003). Cognitive lock-in and the power law of practice. *Journal of Marketing, 67*(2), 62-75. https://doi.org/10.1509/jmkg.67.2.62.18615

Kim, J.-Y. (2026). Data portability and interoperability between digital platforms. *Journal of Economics & Management Strategy, 35*(2), 219-232. https://doi.org/10.1111/jems.12643

Klemperer, P. (1987). Markets with consumer switching costs. *Quarterly Journal of Economics, 102*(2), 375-394. https://doi.org/10.2307/1885068

Klemperer, P. (1995). Competition when consumers have switching costs: An overview with applications to industrial organization, macroeconomics, and international trade. *Review of Economic Studies, 62*(4), 515-539. https://doi.org/10.2307/2298075

Krämer, J. (2021). Personal data portability in the platform economy: Economic implications and policy recommendations. *Journal of Competition Law & Economics, 17*(2), 263-308. https://doi.org/10.1093/joclec/nhaa030

Krämer, J., & Stüdlein, N. (2019). Data portability, data disclosure and data-induced switching costs: Some unintended consequences of the General Data Protection Regulation. *Economics Letters, 181*, 99-103. https://doi.org/10.1016/j.econlet.2019.05.015

Lam, W. M. W., & Liu, X. (2020). Does data portability facilitate entry? *International Journal of Industrial Organization, 69*, Article 102564. https://doi.org/10.1016/j.ijindorg.2019.102564

Lazear, E. P. (2009). Firm-specific human capital: A skill-weights approach. *Journal of Political Economy, 117*(5), 914-940. https://doi.org/10.1086/648671

Milgrom, P., & Segal, I. (2002). Envelope theorems for arbitrary choice sets. *Econometrica, 70*(2), 583-601. https://doi.org/10.1111/1468-0262.00296

Monderer, D., & Shapley, L. S. (1996). Potential games. *Games and Economic Behavior, 14*(1), 124-143. https://doi.org/10.1006/game.1996.0044

Murray, K. B., & Häubl, G. (2007). Explaining cognitive lock-in: The role of skill-based habits of use in consumer choice. *Journal of Consumer Research, 34*(1), 77-88. https://doi.org/10.1086/513048

Nilssen, T. (1992). Two kinds of consumer switching costs. *RAND Journal of Economics, 23*(4), 579-589. https://doi.org/10.2307/2555907

Prüfer, J., & Schottmüller, C. (2021). Competing with big data. *Journal of Industrial Economics, 69*(4), 967-1008. https://doi.org/10.1111/joie.12259

Rhee, K.-E. (2014). What types of switching costs to create under behavior-based price discrimination? *International Journal of Industrial Organization, 37*, 209-221. https://doi.org/10.1016/j.ijindorg.2014.09.003

Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences, 20*(9), 676-688. https://doi.org/10.1016/j.tics.2016.07.002

Shi, M. (2013). A theoretical analysis of endogenous and exogenous switching costs. *Quantitative Marketing and Economics, 11*(2), 205-230. https://doi.org/10.1007/s11129-012-9129-4

Viard, V. B. (2007). Do switching costs make markets more or less competitive? The case of 800-number portability. *RAND Journal of Economics, 38*(1), 146-163. https://doi.org/10.1111/j.1756-2171.2007.tb00049.x

Villas-Boas, J. M. (1999). Dynamic competition with customer recognition. *RAND Journal of Economics, 30*(4), 604-631. https://doi.org/10.2307/2556067

Wernerfelt, B. (1985). Brand loyalty and user skills. *Journal of Economic Behavior & Organization, 6*(4), 381-385. https://doi.org/10.1016/0167-2681(85)90005-8

Wohlfarth, M. (2019). Data portability on the internet: An economic analysis. *Business & Information Systems Engineering, 61*(5), 551-574. https://doi.org/10.1007/s12599-019-00580-9